Arc Length and Improper Integrals

ANSWER KEY

Section 1: Arc Length of a Plane Curve

1 State the formula for the arc length L of $y = f(x)$ from $x = a$ to $x = b$.

a. $L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$ ✓

b. $L = \int_a^b \sqrt{1 + f'(x)}\,dx$

c. $L = \int_a^b [1 + f'(x)]\,dx$

d. $L = \int_a^b \sqrt{[f(x)]^2 + [f'(x)]^2}\,dx$

From $ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + (dy/dx)^2}\,dx$, arc length is $\int_a^b \sqrt{1 + [f'(x)]^2}\,dx$.

2 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

a. $\frac{14}{3}$ ✓

b. 7

c. $\frac{16}{3}$

d. 8

$y' = x^{1/2} \Rightarrow 1 + (y')^2 = 1 + x$. $L = \int_0^3 \sqrt{1 + x}\,dx = [\frac{2}{3}(1 + x)^{3/2}]_0^3 = \frac{2}{3}(4^{3/2} - 1^{3/2}) = \frac{2}{3}(8 - 1) = \frac{14}{3}$.

3 Find the arc length of the line segment $y = \frac{4}{3}x + 2$ from $x = 0$ to $x = 3$.

a. 5 ✓

b. 4

c. 6

d. $\sqrt{7}$

$y' = 4/3 \Rightarrow \sqrt{1 + 16/9} = \sqrt{25/9} = 5/3$. $L = \int_0^3 \frac{5}{3}\,dx = \frac{5}{3}(3) = 5$.

Section 2: Improper Integrals of Type I (Infinite Bounds)

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4 Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$.

a. 1 ✓

b. ∞ (diverges)

c. $1/2$

d. 2

$\lim_{b\to\infty}\int_1^b x^{-2}\,dx = \lim_{b\to\infty}[-1/x]_1^b = \lim_{b\to\infty}(-1/b+1) = 1.$

5 Evaluate $\int_0^\infty e^{-2x}\,dx$.

a. $1/2$ ✓

b. 1

c. ∞

d. 2

$\lim_{b\to\infty}[-\tfrac{1}{2}e^{-2x}]_0^b = \lim_{b\to\infty}(-\tfrac{1}{2}e^{-2b}+1/2) = 1/2.$

6 For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p}\,dx$ converge?

a. $p>1$ ✓

b. $p\ge 1$

c. $p<1$

d. $p>0$

By the p-integral test on $[1,\infty)$, $\int_1^\infty x^{-p}\,dx$ converges if and only if $p>1$. For $p\le 1$, it diverges.

Section 3: Improper Integrals of Type II (Discontinuous Integrands)

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7 Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

a. 2 ✓

b. 1

c. ∞ (diverges)

d. 1/2

Integrand undefined at $x = 0$. $\lim_{a\to 0^+} [2\sqrt{x}]_a^1 = \lim_{a\to 0^+} (2 - 2\sqrt{a}) = 2$.

8 Evaluate $\int_0^2 \frac{1}{x-1}\,dx$.

a. **Diverges** ✓

b. 0

c. $\ln 2$

d. ∞

Infinite discontinuity at $x = 1$. Split: $\int_0^1 + \int_1^2$. $\int_0^1 \frac{dx}{x-1} = \lim_{b\to 1^-} [\ln|x-1|]_0^b = \lim_{b\to 1^-} \ln|b-1| - 0 = -\infty$. Since one part diverges, the integral diverges.

9 For what values of p does $\int_0^1 \frac{1}{x^p}\,dx$ converge?

a. $p < 1$ ✓

b. $p > 1$

c. $p \le 1$

d. $p > 0$

Near 0, $\int_0^1 x^{-p}\,dx = [\frac{x^{1-p}}{1-p}]_0^1$. This converges if and only if the exponent $1 - p > 0 \Rightarrow p < 1$.

Section 4: Comparison Test for Improper Integrals

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10 Determine the convergence of $\int_1^\infty \frac{\sin^2(x)}{x^2}\,dx$.

a. **Converges by comparison with $\frac{1}{x^2}$** ✓

b. Diverges

c. Oscillates without limit

d. Converges to 0

Since $0 \le \frac{\sin^2(x)}{x^2} \le \frac{1}{x^2}$ and $\int_1^\infty \frac{1}{x^2}\,dx$ converges, by the Direct Comparison Test the integral converges.

11 Determine the convergence of $\int_1^\infty \frac{1}{\sqrt{x^3+1}}\,dx$.

a. **Converges (behaves like $x^{-3/2}$ as $x \to \infty$)** ✓

b. Diverges (behaves like $x^{-1/2}$)

c. Diverges to $-\infty$

d. Cannot be tested

For large x, $\frac{1}{\sqrt{x^3+1}} \approx \frac{1}{x^{3/2}}$. By the Limit Comparison Test with $p = 3/2 > 1$, the integral converges.

12 Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$.

a. π ✓

b. 2π

c. $\pi/2$

d. Diverges

Split at 0: $2\int_0^\infty \frac{1}{1+x^2}\,dx = 2\lim_{b\to\infty}[\arctan x]_0^b = 2(\pi/2 - 0) = \pi$.

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